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Inventor(s)	Warth; Jeffrey M.

MOUNT FOR A MACHINING PLATFORM

Abstract

A mount for a machining platform is disclosed. The mount can comprise a body, a first clamp, a second clamp, and a drive screw rotatably mounted to the body. The first clamp and the second clamp can both comprise a clamp body and a protrusion comprising a shoulder. The drive screw can comprise a first external threaded length threadably engaged with a first internal threaded length of the first clamp. The first external threaded length comprises first threads. The drive screw can further comprise a second external threaded length threadably engaged with a second internal threaded length of the second clamp. The second external threaded length comprises second threads oriented in the opposite direction of the first threads. The shoulders move away from each other upon a rotation of the drive screw.

Inventors: **Warth; Jeffrey M. (Mars, PA)**

Applicant: **Chick Workholding Solutions, Inc.** (Warrendale, PA)

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Background/Summary

SUMMARY

[0001] In various aspects, a mount for a machining platform, the mount comprising a body, a first clamp movable relative to the body, a second clamp movable relative to the body, and a drive screw rotatably mounted to the body, is disclosed. The first clamp comprises a first clamp body defining a first opening comprising a first internal threaded length. The first clamp further comprises a first protrusion comprising a first shoulder. The second clamp comprises a second clamp body defining a second opening comprising a second internal threaded length. The second clamp further comprises a second protrusion comprising a second shoulder. The drive screw comprises a first external threaded length threadably engaged with the first internal threaded length of the first clamp. The first external threaded length comprises first threads. The drive screw further comprises a second external threaded length threadably engaged with the second internal threaded length of the second clamp. The second external threaded length comprises second threads oriented in the opposite direction of the first threads. The first shoulder and the second shoulder move away from each other upon a rotation of the drive screw.

[0002] In various aspects, a mount for a machining platform, the mount comprising a body, a first clamp movable relative to the body, a second clamp movable relative to the body, and a drive screw rotatably mounted to the body, is disclosed The first clamp comprises a first clamp body defining a first threaded opening. The first clamp further comprises a first protrusion comprising a first shoulder. The second clamp comprises a second clamp body defining a second threaded opening. The second clamp further comprises a second protrusion comprising a second shoulder. The first shoulder and the second shoulder face away from each other. The drive screw comprises a plurality of first external right-hand threads threadably engaged with the first threaded opening of the first clamp. The drive screw further comprises a plurality of second external left-hand threads threadably engaged with the second threaded opening of the second clamp. The first shoulder and the second shoulder simultaneously move away from each other upon a rotation of the drive screw.

[0003] In various aspects, a mounting system comprising a body, a first clamp movable relative to the body, a second clamp movable relative to the body, and a drive screw rotatably mounted to the body is disclosed. The first clamp comprises a first clamp body defining a first threaded opening. The first clamp further comprises a first protrusion comprising a first shoulder. The second clamp comprises a second clamp body defining a second threaded opening. The second clamp further comprises a second protrusion comprising a second shoulder. The first shoulder and the second shoulder face away from each other. A plane of symmetry is defined between the first clamp and the second clamp. The drive screw comprises a plurality of first external threads threadably engaged with the first threaded opening of the first clamp. The drive screw further comprises a plurality of second external threads threadably engaged with the second threaded opening of the second clamp. The first shoulder and the second shoulder move from a first configuration to a second configuration upon a rotation of the drive screw. The first clamp and the second clamp are equidistantly spaced from the plane of symmetry in the first configuration and in the second configuration.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0004] The features of various aspects are set forth with particularity in the appended claims. The various aspects, however, both as to organization and methods of operation, together with further objects and advantages thereof, may best be understood by reference to the following description, taken in conjunction with the accompanying drawings as follows:

[0005] FIG. **1** is a perspective view of a mount for a machining platform, illustrating several workpiece holders that are compatible with the mount, in accordance with at least one aspect of the present disclosure.

[0006] FIG. **2** is a perspective view of different adapters of varying sizes and configurations, the adapters are compatible with corresponding workpiece holders and the mount of FIG. **1**, in accordance with at least one aspect of the present disclosure.

[0007] FIG. **3** is perspective view of the mount of FIG. **1** attached to a pallet plate of a machining platform, depicting a workpiece holder with an adapter attached thereto positioned between the mount and the workpiece holder, in accordance with at least one aspect of the present disclosure.

[0008] FIG. **4** is a perspective view of the adapter of FIG. **3**, in accordance with at least one aspect of the present disclosure.

[0009] FIG. **5** is another perspective view of the adapter of FIG. **3**, depicting receptacles defined in the adapter for receiving protrusions of the mounting device, in accordance with at least one aspect of the present disclosure.

[0010] FIG. **6** is an exploded view of the mount of FIG. **3**, depicting a base portion, a cover plate, and a pair of clamp assemblies, wherein each clamp assembly includes a pair of clamps and a protrusion for each clamp, in accordance with at least one aspect of the present disclosure.

[0011] FIG. **7** is a perspective view of the different base portions for the mount, in accordance with at least one aspect of the present disclosure.

[0012] FIG. **8** is a perspective view of the mount of FIG. **3** with the cover plate removed, depicting the pair of clamp assemblies attached to the base, in accordance with at least one aspect of the present disclosure.

[0013] FIG. **9** a plan view of the mount of FIG. **3** with the cover plate removed, depicting the pair of clamp assemblies attached to the base, in accordance with at least one aspect of the present disclosure.

[0014] FIG. **10** is a perspective view of the pair of clamp assemblies of the mount of FIG. **3**, in accordance with at least one aspect of the present disclosure.

[0015] FIG. **11** is a perspective view of one of the clamp assemblies of FIG. **8**, in accordance with at least one aspect of the present disclosure.

[0016] FIG. **12** is another perspective view of the clamp assembly of FIG. **11** in accordance with at least one aspect of the present disclosure.

[0017] FIG. **13** is a cross section view of the clamp assembly of FIG. **11**, in accordance with at least one aspect of the present disclosure.

[0018] FIG. **14** is a side elevation view of a drive screw of the clamp assembly of FIG. **11**, in accordance with at least one aspect of the present disclosure.

[0019] FIG. **15** is cross section view of the workpiece holder, the adapter, and the mount of FIG. **3**, depicting the protrusions of the mount positioned within receptacles of the adapter, and further depicting the clamps in a first configuration, in accordance with at least one aspect of the present disclosure.

[0020] FIG. **16** is cross section view of the workpiece holder, the adapter, and the mount of FIG. **3**, depicting the clamps of the mount positioned within receptacles of the adapter, and further depicting the clamps in a second configuration, in accordance with at least one aspect of the present disclosure.

[0021] FIG. **17** is a perspective view of two different adapters, each adapter having a blank surface for customization, in accordance with at least one aspect of the present disclosure.

[0022] FIG. **18** is a perspective view of two different adapters, each adapter having a plurality of mounting holes defined therein, in accordance with at least one aspect of the present disclosure.

[0023] FIG. **19** is a perspective view of the mount of FIG. **1**, depicting the mount attached to a T-slotted machining table, and further depicting an adapter positioned above the mount, in accordance with at least one aspect of the present disclosure.

[0024] FIG. **20** is a perspective view of a mount, depicting the mount mounted to a pallet plate on an indexer with a workpiece holder positioned above the mount, in accordance with at least one aspect of the present disclosure.

[0025] FIG. **21** is a perspective view of the mount of FIG. **20**, depicting the mount mounted to the indexer with the workpiece holder mounted to the mount, in accordance with at least one aspect of the present disclosure.

DETAILED DESCRIPTION

[0026] Before explaining various aspects of the mount in detail, it should be noted that the illustrative examples are not limited in application or use to the details of construction and arrangement of parts illustrated in the accompanying drawings and description. The illustrative examples may be implemented or incorporated in other aspects, variations, and modifications, and may be practiced or carried out in various ways. Further, unless otherwise indicated, the terms and expressions employed herein have been chosen for the purpose of describing the illustrative examples for the convenience of the reader and are not for the purpose of limitation thereof. Also, it will be appreciated that one or more of the following-described aspects, expressions of aspects, and/or examples, can be combined with any one or more of the other following-described aspects, expressions of aspects and/or examples.

[0027] In general, milling machines are used within the metal working industry. Milling machines can utilize removable pallets as a platform to mount materials or fixtures. Installing workpiece holders and fixtures to the milling machines can consume significant set-up time and can require a special set of skills. Various systems in the industry for mounting workpiece holders are referred to as “Zero Point Setting” systems. These systems are limited by their strength and rigidity in many applications. For example, most of these systems are designed to grasp one or more locating pins. The mount and/or mounting systems described herein provide for quick change, accurate fixture installation with a high clamping force and provide a rigid connection. In at least one aspect, the high rigidity allows larger fixtures (e.g., workpiece holders) to be installed to machining platforms.

[0028] FIG. **1** illustrates a mounting system **1000** in which multiple workpiece holders are compatible with a mount **2000**. In at least one aspect, the mount **2000** is attached to a machining platform, such as a milling table, by way of a pallet plate **4000**. Exemplary workpiece holders include a four-sided workpiece holder **1100** having two clamping assemblies on each side forming an eight station holder, a four-sided workpiece holder **1200** having one clamping assembly on each side forming a four station holder, a six-sided workpiece holder **1300** having one clamping assembly on each side forming a six station holder, a four-sided workpiece holder **1400** having two clamping assemblies on each side forming an alternative eight station holder, and a four-sided workpiece holder **1500** having two clamping assemblies on two of the sides and one clamping assembly on the other two sides. Alternative workpiece holders are also contemplated. For example, the workpiece holders **1100**, **1200**, **1300**, **1400**, and **1500** may include a pair of vertically-stacked clamping assemblies on one or more sides. In various instances, more than two clamp assemblies can be stacked on one or more sides of a workpiece holder. In various instances, the workpiece holder can be a MultiLOK workpiece holder from Chick Workholding Solutions, Warrendale, PA. The system **1000** can also include workpiece holders by other manufacturers, which can be compatible with the mount **2000**.

[0029] FIG. **2** illustrates a plurality of adapters **3000** of varying sizes and configurations. In various aspects, the adapters **3000** are compatible with one or more of the corresponding workpiece holders and the mount **2000**. For example, adapter **3200** and adapter **3202** are compatible with the workpiece holder **1200** illustrated in FIG. **1**. Each of the adapters **3000** is shown in a standard height and a raised-height configuration. For example, the adapter **3200** is a standard height adapter and the adapter **3202** is a raised-height adapter. Adapters **3200**, **3202**, **3204**, **3206**, **3208**, **3210**, **3212**, **3214**, **3216**, **3218**, **3220**, **3222**, **3224**, **3226**, **3228**, and **3230** are shown; however, alternative adapters are contemplated. Each adapter is attachable to its respective workpiece holder (e.g. workpiece holders **1100**, **1200**, **1300**, **1400**, **1500** in FIG. **1**) by way of bolts, screws, and/or any other suitable mechanical fastener.

[0030] FIG. **3** illustrates the mount **2000** attached to the pallet plate **4000** and also shows the adapter **3200** attached to the workpiece holder **1200** and positioned above the mount **2000**. In various aspects, when any one of the adapters **3000** of FIG. **2** is attached to a compatible workpiece holder, the resulting assembly may be referred to as a payload. For example, the adapter **3200** attached to the workpiece holder **1200** forms a payload **5000**.

[0031] Referring again to FIG. **2**, each of adapters **3000** comprises receptacles defined therein, which facilitate attachment of the adapter to the mount **2000**. For example, FIGS. **4** and **5** illustrate the adapter **3200** comprising an adapter body **3250** defining a plurality of receptacles **3270**. In at least one aspect, the adapter body **3250** defines four receptacles **3270** for receiving four protrusions of the mount **2000**, as described in greater detail herein. Further, in at least one aspect, the adapter body **3250** defines a recess **3272** within each receptacle **3270** forming a boss **3274** along the rim of the recess **3272**. The recess **3272** is dimensioned and positioned to receive a shoulder of the protrusion positioned therein and the boss **3274** is dimensioned and positioned to retain the shoulder within the recess **3272**, as described in greater detail herein.

[0032] All of the adapters **3000** illustrated in FIG. **2** comprise receptacles defined therein, similar to the receptacles **3270** (FIG. **5**), to facilitate coupling of the adapters **3000** to the mount **2000**. In various aspects, an adapter other than those shown in FIG. **2** may be manufactured or otherwise obtained to adapt another type of workpiece holder to be compatible with the mount **2000**. In at least one aspect, a workpiece holder may comprise the receptacles **3270** without the need for an adapter plate. In various aspects, any adapter and/or workpiece holder can be compatible with the mount **2000** as long as the adapter/workpiece holder comprises the compatible receptacles (e.g. receptacles **3270**) described herein.

[0033] The mount **2000** is depicted in an exploded configuration in FIG. **6** to expose various internal components thereof. The mount **2000** comprises a base **2100**, a cover plate **2200**, a pair of identical slide assemblies, a first clamp assembly **2300** and a second clamp assembly **2400**. Each clamp assembly **2300**, **2400** includes a pair of clamps and a protrusion for each clamp that is configured to be received in the receptacles **3270** of one of the adapters **3000**, such as the adapter **3200**, as described in greater detail below. As further described herein, the mount **2000** is designed such that the base **2100** is first fastened to the machining center or table (e.g. bolted) and the internal components (e.g. clamp assemblies **2300**, **2400**) are then installed and secured via the cover plate **2200**. In such instances, the cover plate **2200** can be removed to access the internal components for service, quality control, and/or maintenance activities without requiring realignment of the base **2100** with the machining center.

[0034] Referring still to FIG. **6**, the base **2100** comprises a base body **2110** defining a first cavity **2120** and a second cavity **2130**. In at least one aspect, the first cavity **2120** and the second cavity **2130** are symmetric about a centerline therebetween and parallel to one another. The first cavity **2120** is to receive the first clamp assembly **2300** and the second cavity **2130** is to receive the second clamp assembly **2400**. The first cavity **2120** comprises a first opening **2122**, a second opening **2124** connected to the first opening **2122**, and a first drive screw mount opening **2126** connected to the second opening **2124**. Further, the second cavity **2130** comprises a first opening **2132**, a second opening **2134** connected to the first opening **2132**, and a second drive screw mount opening **2136** connected to the second opening **2134**. In at least one aspect, the first openings **2122**, **2132** are wider, i.e., laterally larger, than the second openings **2124**, **2134**. The drive screw mount openings **2126**, **2136** extend through a side **2112** of the base body **2110**. Further, the base **2100** further comprises a pair of locating pins **2140** and four reference pins **2142** extending from the base body **2110** for locating and aligning the cover plate **2200** to the base **2100**. In various instances, one of the locating pins **2140** can comprise a first perimeter geometry (e.g. a diamond pin), and the other of the locating pins **2140** can comprise a second perimeter geometry (e.g. a round pin). The locating pins **2140** can be dimensioned and positioned to accurate locate the payloads to the base **2100**, for example.

[0035] In other instances, one or more of the alignment pins and/or reference pins can extend from the cover plate **2200** for mating alignment with geometries on the base **2100**. The base **2100** further comprises a plurality of threaded openings **2160** for receiving screws. Once the first clamp assembly **2300** and the second clamp assembly **2400** are installed to the base **2100**, the cover plate **2200** can be attached to the base **2100**, as described in greater detail below.
[0036] Referring still to FIG. **6**, the cover plate **2200** comprises a plate body **2210** defining a plurality of apertures **2220**, a pair of location pin holes **2230**, and four screw holes **2240**. The pair of location pin holes **2230** are to receive the pair of locating pins **2140** when the cover plate **2200** is installed onto the base **2100**, as shown in FIG. **3**. The four reference pins **2142** are to be received in reference pin holes defined on the underside of the plate body **2210**. As such, the pair of locating pins **2140** and the four reference pins **2142** properly locate and align the cover plate **2200** with the base **2100**. Further, the screw holes **2240** are to align with the threaded openings **2160** when the cover plate **2200** is placed onto the base **2100**. Screws can be threaded into the threaded openings **2160** of the base to attach the cover plate **2200** to the base **2100**. It should be understood that the cover plate **2200** may be attached to the base **2100** by any suitable mechanical fastening means, such as pins, screws, etc., and combinations thereof. In any event, when the cover plate **2200** is attached to the base **2100** with the clamp assemblies **2300**, **2400** installed to the base, the plurality of apertures **2220** of the cover plate **2200** are to receive at least a portion of the clamp assemblies **2300**, **2400**, as shown in FIG. **3**. A portion of each clamp assembly **2300**, **2400** extends through respective apertures **2220** and is exposed and accessible outside the cover plate **2200**.
[0037] Further to the above, FIG. **7** illustrates different bases **2102**, **2104**, **2106** and the base **2100** that may be used as a base for the mount **2000**. Each of the bases **2102**, **2104**, **2106** comprises a base body defining a first cavity dimensioned and structured to receive the first clamp assembly **2300**, and a second cavity dimensioned and structured to receive the second clamp assembly **2400**, similar to the first cavity **2120** and the second cavity **2130** of the base **2100**. Each of the bases **2100**, **2102**, **2104**, **2106** defines different amounts and/or patterns of mounting holes **2150** therein for mounting the mount **2000** to different fixture platforms and/or machining tables, in accordance with various aspects of the present disclosure.
[0038] Referring to FIGS. **8-10**, the first clamp assembly **2300** comprises a first clamp **2310**, a second clamp **2320**, a first drive screw **2330**, and a first drive screw mount **2340**. The first clamp **2310** and the second clamp **2320** are threadably engaged with the first drive screw **2330**. To install the first clamp assembly **2300** to the base **2100**, the first clamp assembly **2300** is positioned in the first cavity **2120** of the base **2100** and the first drive screw mount **2340** is positioned in the first drive screw mount opening **2126**, as shown in FIG. **8**. In at least one aspect, the first drive screw mount **2340** is attached to the base **2100** by a pair of screws **2342**. In any event, the first drive screw **2330** is rotatably secured to the first drive screw mount **2340** and, thus, when the first clamp assembly **2300** is attached to the base **2100**, the first drive screw **2330** is rotatable relative to the base **2100**.
[0039] Further to the above, the second clamp assembly **2400** comprises a third clamp **2410**, a fourth clamp **2420**, a second drive screw **2430**, and a second drive screw mount **2440**. The third clamp **2410** and the fourth clamp **2420** are threadably engaged with the second drive screw **2430**. To install the second clamp assembly **2400** to the base **2100**, the second clamp assembly **2400** is positioned in the second cavity **2130** of the base **2100** and the second drive screw mount **2440** is positioned in the second drive screw mount opening **2136** as shown in FIG. **8**. In at least one aspect, the second drive screw mount **2440** is attached to the base **2100** by a pair of screws **2442**. In any event, the second drive screw **2430** is rotatably secured to the second drive screw mount **2440** and, thus, when the second clamp assembly **2400** is attached to the base **2100**, the second drive screw **2430** is rotatable relative to the base **2100**.
[0040] Referring still to FIGS. **8** and **9**, the second clamp assembly **2400** is spaced apart from the first clamp assembly **2300** and the first and second drive screws **2330**, **2430** extend along parallel

longitudinal axes LA (e.g. FIG. **9**) through the base **2100**. The first clamp assembly **2300** is described in greater detail below. The first and second clamp assemblies **2300**, **2400** are identical. For the sake of brevity, the identical components and features of the second clamp assembly **2400** are not repeated herein.

[0041] Referring now to FIGS. **11-14**, the first clamp assembly **2300** is shown. The first clamp **2310** of the first clamp assembly **2300** comprises a first clamp body **2312** and a first protrusion **2314**. The first clamp body **2312** defines a first opening **2316** comprising a first internal threaded length **2318** (see FIG. **13**). Further, the first protrusion **2314** comprises a first shoulder **2315** and a first groove **2317** positioned underneath the first shoulder **2315**. In at least one aspect, the first shoulder **2315** and the first groove **2317** define a first hook **2319**.

[0042] Further to the above, the second clamp **2320** of the first clamp assembly **2300** comprises a second clamp body **2322** and a second protrusion **2324**. The second clamp body **2322** defines a second opening **2326** comprising a second internal threaded length **2328** (see FIG. **13**). Further, the second protrusion **2324** comprises a second shoulder **2325** and a second groove **2327** positioned underneath the second shoulder **2325**. In at least one aspect, the second shoulder **2325** and the second groove **2327** define a second hook **2329**. In at least one aspect, the first shoulder **2315** and the second shoulder **2325** face opposite directions. In at least one aspect, the first shoulder **2315** and the second shoulder **2325** face away from each other.

[0043] Referring to FIGS. **13** and **14**, the first drive screw **2330** of the first clamp assembly **2300** defines a longitudinal axis LA. The first drive screw **2330** comprises a first external threaded length **2332** threadably engaged with the first internal threaded length **2318** of the first clamp **2310**. The first external threaded length **2332** comprises a plurality of first threads **2333**. Further, the first drive screw **2330** comprises a second external threaded length **2334** threadably engaged with the second internal threaded length **2328** of the second clamp **2320**. The second external threaded length **2334** comprises a plurality of second threads **2335**. In at least one aspect, the second threads **2335** are oriented in the opposite direction of the first threads **2333** such that a rotation of the first drive screw **2330** moves the first clamp **2310** and the second clamp **2320** in opposite directions along the longitudinal axis LA.

[0044] Further to the above, in at least one aspect, the first threads **2333** are right-hand threads and the second threads **2335** are left-hand threads. In at least one aspect, the first threads **2333** are left-hand threads and the second threads **2335** are right-hand threads. Further, in at least one aspect, the first drive screw **2330** defines the first threads **2333** at a first angle FA relative to the longitudinal axis LA and defines the second threads **2335** at a second angle SA relative to the longitudinal axis LA. In at least one aspect, the first angle FA is an acute angle and the second angle SA is an obtuse angle as shown in FIG. **14**. In various instances, the first angle FA and the second angle SA can be supplementary angles, for example.

[0045] Further to the above, in at least one aspect, the first clamp **2310** and the second clamp **2320** are symmetrical about a first plane of symmetry FPS (see FIG. **13**) positioned intermediate the first clamp **2310** and the second clamp **2320**. In at least one aspect, the first clamp **2310** and the second clamp **2320** are identical to one another with the first shoulder **2315** of the first clamp **2310** facing a first direction FD and the second shoulder **2325** of the second clamp **2320** facing a second direction SD that is opposite the first direction FD, see FIG. **13**. In at least one aspect, the first plane of symmetry FPS is orthogonal to the longitudinal axis LA of the first drive screw **2330**.

[0046] Further to the above, in at least one aspect, the first shoulder **2315** defines a first oblique plane FOP transecting the longitudinal axis LA and the second shoulder defines a second oblique plane SOP transecting the longitudinal axis LA. In at least one aspect, the first oblique plane FOB and the second oblique plane SOB are mirrored about the first plane of symmetry FPS. As described in greater detail below, the first clamp **2310** and the second clamp **2320** are configured to move simultaneously relative to one another, relative to the first drive screw **2330**, and relative to the first plane of symmetry FPS upon rotation of the first drive screw **2330**.

[0047] In at least one aspect, when the first clamp assembly **2300** is installed to the base **2100**, as shown in FIG. **8** and as described above, a first rotation of the first drive screw **2330** in a first rotational direction FRD (see FIG. **8**) will cause the first clamp **2310** and the second clamp **2320** to move away from each other. The opposing movement of the clamps **2310**, **2320** is in part due to the first threads **2333** and the second threads **2335** of the first drive screw **2330** being threaded in opposite directions. Moreover, the first clamp **2310** and the second clamp **2320** are rotatably constrained within the first cavity **2120** of the base **2100** such that when the first drive screw **2330** is rotated in the first rotational direction FRD, the first clamp **2310** and the second clamp **2320** are translated away from each other along the longitudinal axis LA.
[0048] Further to the above, in at least one aspect, when the first clamp assembly **2300** is installed to the base **2100**, as shown in FIG. **8** and as described above, a second rotation of the first drive screw **2330** in a second rotational direction SRD that is opposite the first rotational direction FRD will cause the first clamp **2310** and the second clamp **2320** to move toward each other. As discussed above, the opposite threads and rotatably constrained clamps **2310**, **2320** permit the first clamp **2310** and the second clamp **2320** to translate toward each other along the longitudinal axis LA when the first drive screw **2330** is rotated in the second rotational direction SRD. In at least one aspect, no matter the direction or magnitude of rotation of the first drive screw **2330**, the first clamp **2310** and the second clamp **2320** are equidistantly spaced from the first plane of symmetry FPS. Further, in at least one aspect, the first shoulder **2315** and the second shoulder **2325** move away from each other when the first drive screw **2330** is rotated in the first rotational direction FRD and move toward each other when the first drive screw **2330** is rotated in the second rotational direction SRD.
[0049] FIG. **15** illustrates a portion of the payload **5000** (e.g., the adapter **3200** and a portion of the workpiece holder **1200**) positioned onto the mount **2000** with the protrusions **2314**, **2324** of the first and second clamps **2310**, **2320** received within the receptacles **3270** of the adapter **3200**. In at least one aspect, the first clamp **2310** and the second clamp **2320** are positioned the same first distance D**1** from the first plane of symmetry FPS. Further, the first clamp **2310** and the second clamp **2320** are in a first configuration when they are positioned the first distance D**1** from the first plane of symmetry FPS. In at least one aspect, the first configuration of the clamps **2310**, **2320** facilitates placement of the payload **5000** onto the mount **2000** by providing clearance for the protrusions **2314**, **2324** to enter the receptacles **3270**.
[0050] In use, upon rotation of the first drive screw **2330** in the first rotational direction FRD, the first clamp **2310** and the second clamp **2320** move away from the first configuration shown in FIG. **15** and move away from each other to a second configuration shown in FIG. **16**. As the first clamp **2310** and the second clamp **2320** move from the first configuration to the second configuration, the first shoulder **2315** and the second shoulder **2325** engage the corresponding recesses **3272** of the receptacles **3270** of the adapter **3200**, as shown in FIG. **16**. More specifically, the first hook **2319** is structured to hook around the boss **3274** and extend into the recess **2372** of one of the receptacles **3270** to secure the first clamp **2310** to the adapter **3200**/workpiece **1200**, and the second hook **2329** is structured to hook around the boss **3274** and extend into the recess **3272** of another receptacle **3270** to clamp or lock the second claim **2320** to the adapter **3200**/workpiece **1200**. Further, the first clamp **2310** and the second clamp **2320** are positioned the same second distance D**2** from the first plane of symmetry FPS when the first clamp **2310** and the second clamp **2320** are in the second configuration, see FIG. **16**. As discussed above, no matter the direction or magnitude of rotation of the first drive screw **2330**, the first clamp **2310** and the second clamp **2320** are equidistantly spaced from the first plane of symmetry FPS.
[0051] Referring to FIG. **16**, when first clamp **2310** and the second clamp **2320** are in the second configuration, the first clamp **2310** applies a first force F**1** to the receptacle **3270** of the adapter **3200** and the second clamp **2320** applies a second force F**2** to the receptacle **3270** of the adapter **3200**. In at least one aspect, the first force F**1** and the second force F**2** are equal and opposite. In at

least one aspect, the first force F**1** is orthogonal to the first oblique plane FOB (see FIG. **13**) and the second force F**2** is orthogonal to the second oblique plane (see FIG. **13**). In at least one aspect, the forces F**1** and F**2** are applied at an angle relative to the first plane of symmetry FPS. As such, the applied forces F**1** and F**2** each have an outward component and a downward component and thus, the clamps **2310**, **2320** press outward on and pull downward on the adapter **3200** to retain the payload **5000** to the mount **2000**.

[0052] Referring to FIGS. **8** and **9**, both of the first clamp assembly **2300** and the second clamp assembly **2400** are shown installed to the base **2100**, as described above. The first drive screw **2330** of the first clamp assembly **2300** comprises a grip **2337** (e.g., a hex head) defined at an end thereof. The grip **2337** is rotatable by a user with a tool (e.g., a socket) to rotate the first drive screw **2330** relative to the base **2100** of the mount **2000**. Similarly, the second drive screw **2430** of the second clamp assembly **2400** comprises a grip **2437** (e.g., a hex head) defined at an end thereof. The grip **2437** is rotatable by a user with a tool (e.g., a socket) to rotate the second drive screw **2430** relative to the base **2100** of the mount **2000**. The first drive screw **2330** and the second drive screw **2430** are separately actuatable. In at least one aspect the first drive screw **2330** and the second drive screw **2430** can be actuated simultaneously.

[0053] Referring still to FIGS. **8** and **9**, in at least one aspect, the third clamp **2410** and the fourth clamp **2420** of the second clamp assembly **2400** are symmetrical about a second plane of symmetry SPS (see FIG. **9**). In at least one aspect, when the first clamp assembly **2300** and the second clamp assembly **2400** are installed to the base **2100**, the first plane of symmetry FPS and the second plane of symmetry SPS are aligned. In other words, the first plane of symmetry FPS and the second plane of symmetry SPS are co-planar. Further, in at least one aspect, the first clamp assembly **2300** and the second clamp assembly **2400** are symmetrical about a third plane of symmetry TPS, see FIG. **9**. In at least one aspect, when the clamp assemblies **2300**, **2400** are installed to the base **2100**, the third plane of symmetry TPS is parallel to the longitudinal axis LA of the first drive screw **2330** and the longitudinal axis LA of the second drive screw **2430**. Moreover, in at least one aspect, the first plane of symmetry FPS and the second plane of symmetry SPS are orthogonal to the third plane of symmetry TPS.

[0054] Further to the above, it should be understood that, in various aspects, the second clamp assembly **2400** is identical structurally and functionally to the first clamp assembly **2300** such that the third clamp **2410** and the fourth clamp **2420** of the second clamp assembly **2300** (see FIG. **8**) engage respective receptacles **3270** of the adapter **3200** in the same manner as the first clamp **2310** and the second clamp **2320** of the first clamp assembly **2300**, as described above.

[0055] As discussed above, the mount **2000** can be attached to a workpiece holder, by way of an adapter, such as the adapters shown in FIG. **2**. Further, FIG. **17** illustrates adapters **3300**, **3400** that are blank to allow customization. For example, any hole/aperture pattern could be made in the adapters **3300**, **3400** to facilitate attachment of adapters **3300**, **3400** to various types of workpiece holders and/or to facilitate attachment of the mount **2000** to the workpiece holders. Further, FIG. **18** illustrates adapters **3500** and **3600** having holes **3550**, **3650** defined therein to facilitate attachment of the adapters **3500**, **3600** to a workpiece holder and/or to facilitate attachment of the mount **2000** to the adapters **3500**, **3600**, for example. In various aspects, the adapters **3300**, **3400**, **3500**, **3600** can be customized to comprise the receptacles (e.g. receptacles **3270**) described herein to facilitate compatibility with the mount **2000**.

[0056] As discussed herein, the mount **2000** is compatible with various types of machining platforms and may or may not require a pallet plate, such as the pallet plate **4000**, to be attached to a machining platform. For example, For example, FIG. **19** illustrates the mount **2000** attached to a T-slotted machining platform **6000**. The machining platform **6000** comprises a body portion **6100** defining a plurality of longitudinal T-slots **6200**. Fasteners such as screws, bolts, etc., and combinations thereof, may be utilized to attach the base **2100** of the mount **2000** directly to the T-slotted machining table **6000**. The machining platform **6000** can be oriented horizontally or

vertically. For example, a T-slotted machining platform is typical for various vertical machining centers.
[0057] Further to the above, FIG. **20** illustrates a mount **7000** similar to the mount **2000** except for the difference described herein. The mount **7000** comprises a base **7100** having a partially circular or arcuate perimeter. The base **7100** may be similar to the base **2100**. For example, in at least one aspect, the base **7100** comprise the first cavity **2120** and the second cavity **2130** of the base **2100** (see FIG. **6**) for receiving the first clamp assembly **2300** and the second clamp assemblies **2400**, respectively. In any event, the base **7100** is mounted to a pallet plate **4800** that is mechanically attached to an indexer, or 4-axis machining table, **8000**. Once the base **7100** is attached to the pallet plate **4800**, a workpiece holder **9000** can be mounted to the mount **7000**, as described herein in relation to the mount **2000**. In at least one aspect, the workpiece holder **9000** must first be outfitted with an adapter, such as one of the adapters **3000** shown in FIG. **2**, in order to be mounted to the mount **7000**. In at least one aspect, the workpiece holder **9000** comprises receptacles defined directly into the bottom of the workpiece holder **9000** to facilitate mounting of the workpiece holder **9000** onto the mount **7000** without a separate adapter component. In at least one aspect, the receptacles defined directly into the workpiece holder **9000** may be similar to the receptacles **3270** of the adapter **3200**.
[0058] The reader will appreciate that the various mounts disclosed herein can be utilized with horizontal and vertical machining centers and with or without indexers, for example. The mounts disclosed herein facilitate a quick-mount capability with sufficient force to securely hold a payload on the various machining centers. Upon installation of the mounts disclosed herein onto a machining table, a variety of workpiece holders can be quickly, rigidly, and accurately mounted and dismounted to the machining table via the mount.
[0059] Various aspects of the subject matter described herein are set out in the following numbered examples.
[0060] Example 1—A mount for a machining platform, the mount comprising a body, a first clamp movable relative to the body, a second clamp movable relative to the body, and a drive screw rotatably mounted to the body. The first clamp comprises a first clamp body defining a first opening comprising a first internal threaded length. The first clamp further comprises a first protrusion comprising a first shoulder. The second clamp comprises a second clamp body defining a second opening comprising a second internal threaded length. The second clamp further comprises a second protrusion comprising a second shoulder. The drive screw comprises a first external threaded length threadably engaged with the first internal threaded length of the first clamp. The first external threaded length comprises first threads. The drive screw further comprises a second external threaded length threadably engaged with the second internal threaded length of the second clamp. The second external threaded length comprises second threads oriented in the opposite direction of the first threads. The first shoulder and the second shoulder move away from each other upon a rotation of the drive screw.
[0061] Example 2—The mount of Example 1, wherein the first external threaded length is right-handed, and wherein the second external threaded length is left-handed.
[0062] Example 3—The mount of Example 1 or 2, wherein the first shoulder and the second shoulder face opposite directions.
[0063] Example 4—The mount of Examples 1, 2, or 3, wherein the first clamp and the second clamp are equidistantly spaced from a plane of symmetry positioned intermediate the first clamp and the second clamp.
[0064] Example 5—The mount of Example 4, wherein the first clamp and the second clamp are symmetrical about the plane of symmetry.
[0065] Example 6—The mount of Example 4 or 5, wherein the drive screw defines a longitudinal axis, wherein the first shoulder defines a first oblique plane transecting the longitudinal axis, wherein the second shoulder defines a second oblique plane transecting the longitudinal axis, and

wherein the first oblique plane and the second oblique plane are mirrored about the plane of symmetry.

[0066] Example 7—The mount of Examples 1, 2, 3, 4, 5, or 6, wherein the first protrusion further comprises a first groove, and wherein the first shoulder and the first groove define a hook.

[0067] Example 8—The mount of Examples 1, 2, 3, 4, 5, 6, or 7, wherein the body comprises a cover plate, and wherein the first protrusion and the second protrusion extend through the cover plate.

[0068] Example 9—The mount of Examples 1, 2, 3, 4, 5, 6, 7, or 8, wherein the drive screw comprises a first drive screw, wherein the mount further comprises a third clamp movable relative to the body, a fourth clamp movable relative to the body, and a second drive screw rotatably mounted to the body. The third clamp comprises a third clamp body defining a third opening comprising a third internal threaded length. The third clamp further comprises a third protrusion comprising a third shoulder. The fourth clamp comprises a fourth clamp body defining a fourth opening comprising a fourth internal threaded length. The fourth clamp further comprises a fourth protrusion comprising a fourth shoulder. The second drive screw is parallel to the first drive screw. The second drive screw comprises a third external threaded length threadably engaged with the third internal threaded length of the third clamp. The second drive screw further comprises a fourth external threaded length threadably engaged with the fourth internal threaded length of the fourth clamp. The fourth external threaded length is in the opposite direction of the third external threaded length. The third shoulder and the fourth shoulder move away from each other upon a rotation of the second drive screw.

[0069] Example 10—A mount for a machining platform, the mount comprising a body, a first clamp movable relative to the body, a second clamp movable relative to the body, and a drive screw rotatably mounted to the body. The first clamp comprises a first clamp body defining a first threaded opening. The first clamp further comprises a first protrusion comprising a first shoulder. The second clamp comprises a second clamp body defining a second threaded opening. The second clamp further comprises a second protrusion comprising a second shoulder. The first shoulder and the second shoulder face away from each other. The drive screw comprises a plurality of first external right-hand threads threadably engaged with the first threaded opening of the first clamp. The drive screw further comprises a plurality of second external left-hand threads threadably engaged with the second threaded opening of the second clamp. The first shoulder and the second shoulder simultaneously move away from each other upon a rotation of the drive screw.

[0070] Example 11—The mount of Example 10, wherein a plane of symmetry is defined between the first clamp and the second clamp, wherein the drive screw defines a longitudinal axis orthogonal to the plane of symmetry, wherein the first clamp and the second clamp are equidistantly spaced from the plane of symmetry.

[0071] Example 12—The mount of Example 11, wherein the first shoulder defines a first oblique plane transecting the longitudinal axis, wherein the second shoulder defines a second oblique plane transecting the longitudinal axis, and wherein the first oblique plane and the second oblique plane are mirrored about the plane of symmetry.

[0072] Example 13—The mount of Examples 10, 11, or 12, wherein the first protrusion comprises a first groove, and wherein the first shoulder and the first groove define a hook.

[0073] Example 14—The mount of Examples 10, 11, 12, or 13, wherein the body comprises a cover plate, and wherein the first protrusion and the second protrusion extend through the cover plate.

[0074] Example 15—The mount of Examples 10, 11, 12, 13, or 14, wherein the drive screw comprises a first drive screw, and wherein the mount further comprises a third clamp movable relative to the body, a fourth clamp movable relative to the body, and a second drive screw rotatably mounted to the body. The third clamp comprises a third clamp body defining a third threaded opening. The third clamp further comprises a third protrusion comprising a third shoulder.

The fourth clamp comprises a fourth clamp body defining a fourth threaded opening. The fourth clamp further comprises a fourth protrusion comprising a fourth shoulder. The third shoulder and the fourth shoulder face away from each other. The second drive screw is parallel to the first drive screw. The second drive screw comprises a plurality of third external right-hand threads threadably engaged with the third threaded opening of the third clamp. The second drive screw further comprises a plurality of fourth external left-hand threads threadably engaged with the fourth threaded opening of the fourth clamp. The third shoulder and the fourth shoulder simultaneously move away from each other upon a rotation of the second drive screw.

[0075] Example 16—A mounting system, comprising a body, a first clamp movable relative to the body, a second clamp movable relative to the body, and a drive screw rotatably mounted to the body. The first clamp comprises a first clamp body defining a first threaded opening. The first clamp further comprises a first protrusion comprising a first shoulder. The second clamp comprises a second clamp body defining a second threaded opening. The second clamp further comprises a second protrusion comprising a second shoulder. The first shoulder and the second shoulder face away from each other. A plane of symmetry is defined between the first clamp and the second clamp. The drive screw comprises a plurality of first external threads threadably engaged with the first threaded opening of the first clamp. The drive screw further comprises a plurality of second external threads threadably engaged with the second threaded opening of the second clamp. The first shoulder and the second shoulder move from a first configuration to a second configuration upon a rotation of the drive screw. The first clamp and the second clamp are equidistantly spaced from the plane of symmetry in the first configuration and in the second configuration.

[0076] Example 17—The mounting system of Example 16, wherein the first protrusion comprises a first groove, and wherein the first shoulder and the first groove define a hook.

[0077] Example 18—The mounting system of Example 16 or 17, wherein the drive screw comprises a first drive screw and the plane of symmetry comprises a first plane of symmetry, and wherein the mounting system further comprises a third clamp movable relative to the body, a fourth clamp movable relative to the body, and a second drive screw rotatably mounted to the body. The third clamp comprises a third clamp body defining a third threaded opening. The third clamp further comprises a third protrusion comprising a third shoulder. The fourth clamp comprises a fourth clamp body defining a fourth threaded opening. The fourth clamp further comprises a fourth protrusion comprising a fourth shoulder. The third shoulder and the fourth shoulder face away from each other. A second plane of symmetry is defined between the third clamp and the fourth clamp. The second drive screw is parallel to the first drive screw. The second drive screw comprises a plurality of third external threads threadably engaged with the third threaded opening of the third clamp. The second drive screw comprises a plurality of fourth external threads threadably engaged with the fourth threaded opening of the fourth clamp. The third shoulder and the fourth shoulder move from a third configuration to a fourth configuration upon a rotation of the second drive screw. The third clamp and the fourth clamp are equidistantly spaced from the second plane of symmetry in the third configuration and in the fourth configuration.

[0078] Example 19—The mounting system of Example 18, wherein the mounting system further comprises a payload, wherein the first clamp and the second clamp are positioned to exert equal and opposite clamping forces on the payload in the second configuration, and wherein the third clamp and the fourth clamp are positioned to exert equal and opposite clamping forces on the payload in the fourth configuration.

[0079] Example 20—The mounting system of Example 18 or 19, wherein a third plane of symmetry is defined between the first clamp and the third clamp such that the first clamp and the third clamp are equidistantly spaced from the third plane of symmetry, and wherein the first drive screw and the second drive screw extend longitudinally on opposite sides of the third plane of symmetry.

[0080] While several forms have been illustrated and described, it is not the intention of the applicant to restrict or limit the scope of the appended claims to such detail. Numerous modifications, variations, changes, substitutions, combinations, and equivalents to those forms may be implemented and will occur to those skilled in the art without departing from the scope of the present disclosure. Moreover, the structure of each element associated with the described forms can be alternatively described as a means for providing the function performed by the element. Also, where materials are disclosed for certain components, other materials may be used. It is therefore to be understood that the foregoing description and the appended claims are intended to cover all such modifications, combinations, and variations as falling within the scope of the disclosed forms. The appended claims are intended to cover all such modifications, variations, changes, substitutions, modifications, and equivalents.

[0081] As used in any aspect herein, the terms “component,” “system,” “module” and the like can refer to a computer-related entity, either hardware, a combination of hardware and software, software, or software in execution.

[0082] One or more components may be referred to herein as “configured to,” “configurable to,” “operable/operative to,” “adapted/adaptable,” “able to,” “conformable/conformed to,” etc. Those skilled in the art will recognize that “configured to” can generally encompass active-state components and/or inactive-state components and/or standby-state components, unless context requires otherwise.

[0083] It will be further appreciated that, for convenience and clarity, spatial terms such as “vertical”, “horizontal”, “up”, and “down” may be used herein with respect to the drawings. However, treatment systems can be used in many orientations and positions, and these terms are not intended to be limiting and/or absolute.

[0084] Those skilled in the art will recognize that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to claims containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations.

[0085] In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances where a convention analogous to “at least one of A, B, or C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that typically a disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms unless context dictates otherwise. For example, the phrase "A or B" will be typically understood to include the possibilities of "A" or "B" or "A and B."

[0086] With respect to the appended claims, those skilled in the art will appreciate that recited operations therein may generally be performed in any order. Also, although various operational flow diagrams are presented in a sequence(s), it should be understood that the various operations may be performed in other orders than those which are illustrated, or may be performed concurrently. Examples of such alternate orderings may include overlapping, interleaved, interrupted, reordered, incremental, preparatory, supplemental, simultaneous, reverse, or other variant orderings, unless context dictates otherwise. Furthermore, terms like "responsive to," "related to," or other past-tense adjectives are generally not intended to exclude such variants, unless context dictates otherwise.

[0087] It is worthy to note that any reference to "one aspect," "an aspect," "an exemplification," "one exemplification," and the like means that a particular feature, structure, or characteristic described in connection with the aspect is included in at least one aspect. Thus, appearances of the phrases "in one aspect," "in an aspect," "in an exemplification," and "in one exemplification" in various places throughout the specification are not necessarily all referring to the same aspect. Furthermore, the particular features, structures or characteristics may be combined in any suitable manner in one or more aspects.

[0088] Any patent application, patent, non-patent publication, or other disclosure material referred to in this specification and/or listed in any Application Data Sheet is incorporated by reference herein, to the extent that the incorporated materials is not inconsistent herewith. As such, and to the extent necessary, the disclosure as explicitly set forth herein supersedes any conflicting material incorporated herein by reference. Any material, or portion thereof, that is said to be incorporated by reference herein, but which conflicts with existing definitions, statements, or other disclosure material set forth herein will only be incorporated to the extent that no conflict arises between that incorporated material and the existing disclosure material.

[0089] In summary, numerous benefits have been described which result from employing the concepts described herein. The foregoing description of the one or more forms has been presented for purposes of illustration and description. It is not intended to be exhaustive or limiting to the precise form disclosed. Modifications or variations are possible in light of the above teachings. The one or more forms were chosen and described in order to illustrate principles and practical application to thereby enable one of ordinary skill in the art to utilize the various forms and with various modifications as are suited to the particular use contemplated. It is intended that the claims submitted herewith define the overall scope.

Claims

1. A mount for a machining platform, the mount comprising: a body; a first clamp movable relative to the body, the first clamp comprising: a first clamp body defining a first opening comprising a first internal threaded length; and a first protrusion comprising a first shoulder; a second clamp movable relative to the body, the second clamp comprising: a second clamp body defining a second opening comprising a second internal threaded length; and a second protrusion comprising a second shoulder; and a drive screw rotatably mounted to the body, the drive screw comprising: a first external threaded length threadably engaged with the first internal threaded length of the first clamp, wherein the first external threaded length comprises first threads; and a second external threaded length threadably engaged with the second internal threaded length of the second clamp, wherein the second external threaded length comprises second threads oriented in the opposite direction of the first threads, and wherein the first shoulder and the second shoulder move away from each other upon a rotation of the drive screw.

2. The mount of claim 1, wherein the first external threaded length is right-handed, and wherein the second external threaded length is left-handed.

3. The mount of claim 1, wherein the first shoulder and the second shoulder face opposite directions.

4. The mount of claim 1, wherein the first clamp and the second clamp are equidistantly spaced from a plane of symmetry positioned intermediate the first clamp and the second clamp.

5. The mount of claim 4, wherein the first clamp and the second clamp are symmetrical about the plane of symmetry.

6. The mount of claim 5, wherein the drive screw defines a longitudinal axis, wherein the first shoulder defines a first oblique plane transecting the longitudinal axis, wherein the second shoulder defines a second oblique plane transecting the longitudinal axis, and wherein the first oblique plane and the second oblique plane are mirrored about the plane of symmetry.

7. The mount of claim 1, wherein the first protrusion further comprises a first groove, and wherein the first shoulder and the first groove define a hook.

8. The mount of claim 1, wherein the body comprises a cover plate, and wherein the first protrusion and the second protrusion extend through the cover plate.

9. The mount of claim 1, wherein the drive screw comprises a first drive screw, wherein the mount further comprises: a third clamp movable relative to the body, the third clamp comprising: a third clamp body defining a third opening comprising a third internal threaded length; and a third protrusion comprising a third shoulder; a fourth clamp movable relative to the body, the fourth clamp comprising: a fourth clamp body defining a fourth opening comprising a fourth internal threaded length; and a fourth protrusion comprising a fourth shoulder; and a second drive screw rotatably mounted to the body, wherein the second drive screw is parallel to the first drive screw, the second drive screw comprising: a third external threaded length threadably engaged with the third internal threaded length of the third clamp; and a fourth external threaded length threadably engaged with the fourth internal threaded length of the fourth clamp, wherein the fourth external threaded length is in the opposite direction of the third external threaded length, and wherein the third shoulder and the fourth shoulder move away from each other upon a rotation of the second drive screw.

10. A mount for a machining platform, the mount comprising: a body; a first clamp movable relative to the body, the first clamp comprising: a first clamp body defining a first threaded opening; and a first protrusion comprising a first shoulder; a second clamp movable relative to the body, the second clamp comprising: a second clamp body defining a second threaded opening; and a second protrusion comprising a second shoulder, wherein the first shoulder and the second shoulder face away from each other; and a drive screw rotatably mounted to the body, the drive screw comprising: a plurality of first external right-hand threads threadably engaged with the first threaded opening of the first clamp; and a plurality of second external left-hand threads threadably engaged with the second threaded opening of the second clamp, wherein the first shoulder and the second shoulder simultaneously move away from each other upon a rotation of the drive screw.

11. The mount of claim 10, wherein a plane of symmetry is defined between the first clamp and the second clamp, wherein the drive screw defines a longitudinal axis orthogonal to the plane of symmetry, wherein the first clamp and the second clamp are equidistantly spaced from the plane of symmetry.

12. The mount of claim 11, wherein the first shoulder defines a first oblique plane transecting the longitudinal axis, wherein the second shoulder defines a second oblique plane transecting the longitudinal axis, and wherein the first oblique plane and the second oblique plane are mirrored about the plane of symmetry.

13. The mount of claim 10, wherein the first protrusion comprises a first groove, and wherein the

first shoulder and the first groove define a hook.
14. The mount of claim 10, wherein the body comprises a cover plate, and wherein the first protrusion and the second protrusion extend through the cover plate.
15. The mount of claim 10, wherein the drive screw comprises a first drive screw, and wherein the mount further comprises: a third clamp movable relative to the body, the third clamp comprising: a third clamp body defining a third threaded opening; and a third protrusion comprising a third shoulder; a fourth clamp movable relative to the body, the fourth clamp comprising: a fourth clamp body defining a fourth threaded opening; a fourth protrusion comprising a fourth shoulder, wherein the third shoulder and the fourth shoulder face away from each other; and a second drive screw rotatably mounted to the body, wherein the second drive screw is parallel to the first drive screw, the second drive screw comprising: a plurality of third external right-hand threads threadably engaged with the third threaded opening of the third clamp; and a plurality of fourth external left-hand threads threadably engaged with the fourth threaded opening of the fourth clamp, wherein the third shoulder and the fourth shoulder simultaneously move away from each other upon a rotation of the second drive screw.
16. A mounting system, comprising: a body; a first clamp movable relative to the body, the first clamp comprising: a first clamp body defining a first threaded opening; and a first protrusion comprising a first shoulder; a second clamp movable relative to the body, the second clamp comprising: a second clamp body defining a second threaded opening; and a second protrusion comprising a second shoulder, wherein the first shoulder and the second shoulder face away from each other, wherein a plane of symmetry is defined between the first clamp and the second clamp; and a drive screw rotatably mounted to the body, the drive screw comprising: a plurality of first external threads threadably engaged with the first threaded opening of the first clamp; and a plurality of second external threads threadably engaged with the second threaded opening of the second clamp, wherein the first shoulder and the second shoulder move from a first configuration to a second configuration upon a rotation of the drive screw, and wherein the first clamp and the second clamp are equidistantly spaced from the plane of symmetry in the first configuration and in the second configuration.
17. The mounting system of claim 16, wherein the first protrusion comprises a first groove, and wherein the first shoulder and the first groove define a hook.
18. The mounting system of claim 16, wherein the drive screw comprises a first drive screw and the plane of symmetry comprises a first plane of symmetry, and wherein the mounting system further comprises: a third clamp movable relative to the body, the third clamp comprising: a third clamp body defining a third threaded opening; and a third protrusion comprising a third shoulder; a fourth clamp movable relative to the body, the fourth clamp comprising: a fourth clamp body defining a fourth threaded opening; and a fourth protrusion comprising a fourth shoulder, wherein the third shoulder and the fourth shoulder face away from each other, wherein a second plane of symmetry is defined between the third clamp and the fourth clamp; and a second drive screw rotatably mounted to the body, wherein the second drive screw is parallel to the first drive screw, the second drive screw comprising: a plurality of third external threads threadably engaged with the third threaded opening of the third clamp; and a plurality of fourth external threads threadably engaged with the fourth threaded opening of the fourth clamp, wherein the third shoulder and the fourth shoulder move from a third configuration to a fourth configuration upon a rotation of the second drive screw, and wherein the third clamp and the fourth clamp are equidistantly spaced from the second plane of symmetry in the third configuration and in the fourth configuration.
19. The mounting system of claim 18, wherein the mounting system further comprises a payload, wherein the first clamp and the second clamp are positioned to exert equal and opposite clamping forces on the payload in the second configuration, and wherein the third clamp and the fourth clamp are positioned to exert equal and opposite clamping forces on the payload in the fourth configuration.

20. The mounting system of claim 19, wherein a third plane of symmetry is defined between the first clamp and the third clamp such that the first clamp and the third clamp are equidistantly spaced from the third plane of symmetry, and wherein the first drive screw and the second drive screw extend longitudinally on opposite sides of the third plane of symmetry.